

1. **Figure 1** is a diagram showing the stages in testing leaves. The arrows in the diagram show the sequence of the process. A, B, C, D, E in the diagram are liquids. Use the diagram to answer the questions that follow.

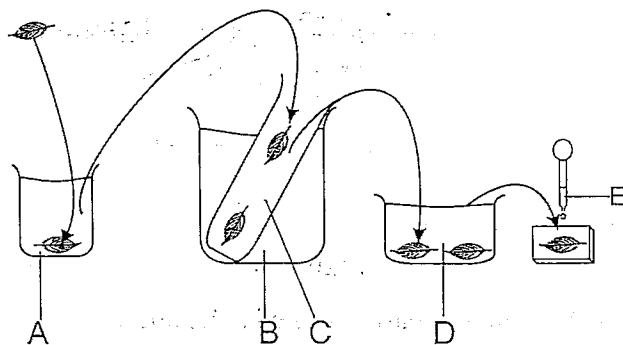


Figure 1

- Which of the liquids is
 - iodine solution?
 - used to kill leaves?
 - used to remove colour from leaves?
- State *two* advantages of killing leaves.
- Why is liquid C normally boiled in liquid B?
- What would be the result of testing for starch a leaf that has been
 - in the dark for 24 hours?
 - in the sun for 4 hours?
 Give reasons for your answers.

2. **Figure 2** is a diagram showing the rib cage cut open to show the internal structures. Use it to answer the questions that follow.

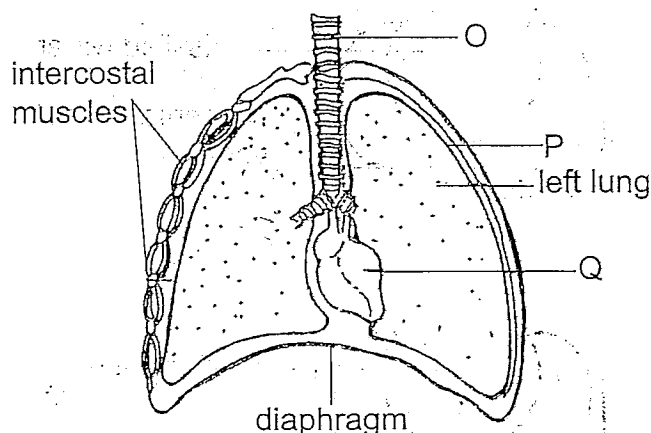


Figure 2

- Name the parts marked O and Q.

- Name the liquid found in the region marked P.
 - What role is played by the fluid in (b)i
- State the role played by each of the following structures during inspiration.
 - Diaphragm
 - Intercostal muscles

3. **Figure 3** is a diagram showing a plant cell. The arrows show the movement of substances in the cell.

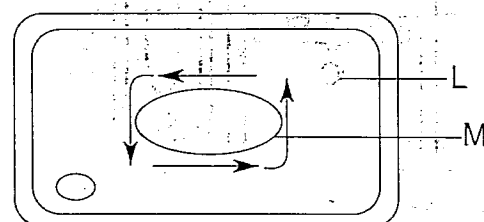


Figure 3

- Name the parts marked L and M.
 - What name is given to movement of substances shown in figure 3?
 - In what way is the movement of substance shown in figure 3
 - similar to active transport?
 - more advantageous to the cell than diffusion?
4. In an investigation on soil composition, a student heated 250 g of garden soil to constant mass using the apparatus shown in **figure 4**. Use it to answer the questions that follow.

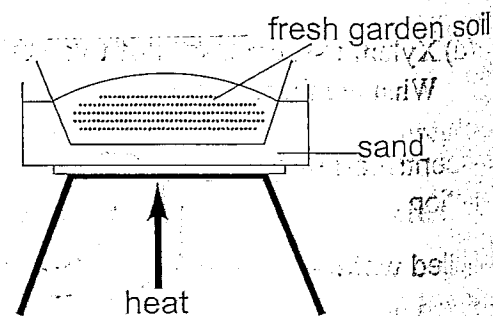


Figure 4

- Describe how the student heated the soil to constant mass.
- What was the aim of the investigation?
- Explain briefly why the garden soil was heated using a sand bath as shown in figure 4.

Figure 5 is a diagram showing the kidney and its associated organs. Use it to answer the questions that follow.

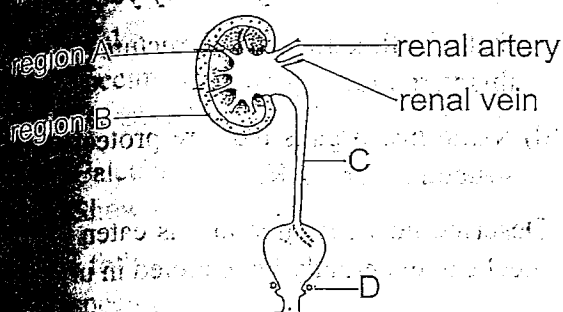


Figure 5

- (a) Name the parts marked C and D.
- (b) In which region is the Bowman's capsule located?
- (c) State *two* differences in the composition between the blood in the two renal vessels shown in figure 5.
6. A student estimated the population of five species of fish in a pond. Table 1 shows the results obtained. Use it to answer the questions that follow.

Table 1

SPECIES	E	F	G	H	I
Number of fish	2250	1000	250	125	400

- (a) i. Draw a bar chart to compare number of fish of each species in the pond.
ii. Which is the least abundant fish species in the pond?
- (b) Apart from fertiliser, mention *two* factors that may affect the growth of fish population.
- (c) How can a fish population in a pond be estimated without counting all the fish?
7. (a) Describe how bacteria and leguminous plants benefit from their symbiotic relationship in a root nodule.
(b) State *two* advantages and *two* disadvantages of being an epiphyte.
8. Figure 6 is a diagram showing two pure lines of peas that Gregor Mendel cross-pollinated in his experiments on heredity. Use it to answer the questions that follow.

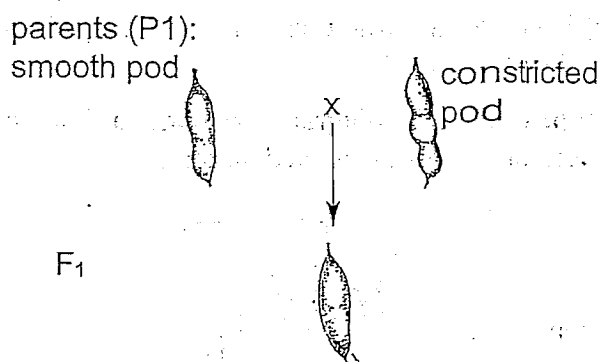


Figure 6

- (a) i. Which of the two pure lines has a recessive characteristic?
ii. Explain your answer to (a)i.
- (b) What is meant by "pure line peas"?
- (c) When the F_1 are selfed, 25% of the F_2 have constricted pods.
i. What is meant by "selfing F_1 "?
ii. Using the symbol B for dominant gene and b for the recessive gene, the gene of the F_1 parents is Bb . Work out the genotype and phenotype of the F_2 generation. Show your working.
9. Figures 7A, 7B, 7C, and 7D are diagrams showing a plant root tip and three cells taken from points 1, 2, 3 on the root tip. Use them to answer the questions that follow.

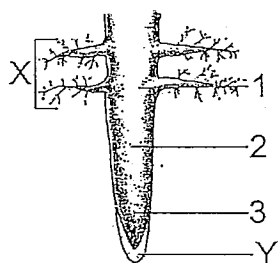


Figure 7A

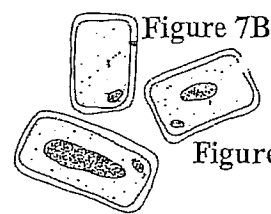


Figure 7C

- (a) Complete the table by matching a cell with the region of the root tip from which it has been taken.

CELL	REGION OF ROOT TIP

- (b) State the function of the structures marked X and Y.

(c) How are the structures marked X adapted to their function?

10. Figure 6 is a diagram showing a human vertebra and its associated structures.

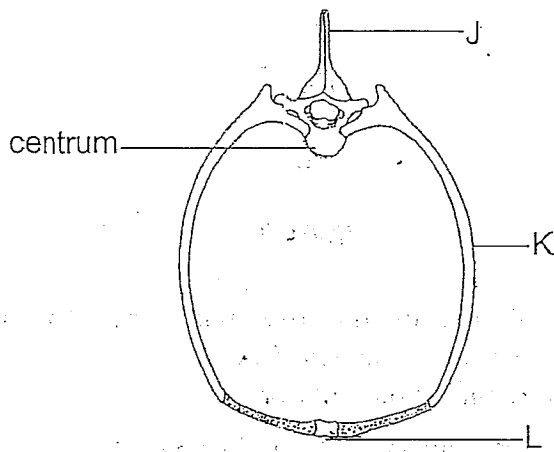


Figure 8

- Name the structures marked K and L.
- To which region of the vertebral column does this vertebra belong? Justify your answer.
- State the function of the structure marked J to this vertebra.
- Name *two* organs that are protected by the structure marked K.

11. Describe how nitrogen atoms eaten in a bean meal can eventually be excreted in urine.

12. Explain how the stem of a plant is adapted to its function.